

Lab 7: Molecular Genetics — From DNA to Protein

BIOL-1

Name: _____ Date: _____

Overview

In this lab, you will walk through the Central Dogma step by step: transcribe a DNA gene into mRNA, then translate the mRNA into a protein using a codon table. You will also explore what happens when mutations change the DNA.

Materials

- This worksheet
- Codon table (provided below)
- Colored pencils (optional)

Part 1: Codon Table

Use this table to translate mRNA codons into amino acids. Find the **first base** on the left, the **second base** on top, and the **third base** on the right.

U	C	A	G
U Phe, Phe, Leu, Leu	Ser, Ser, Ser, Ser	Tyr, Tyr, STOP, STOP	Cys, Cys, STOP, Trp
C Leu, Leu, Leu, Leu	Pro, Pro, Pro, Pro	His, His, Gln, Gln	Arg, Arg, Arg, Arg
A Ile, Ile, Ile, Met (START)	Thr, Thr, Thr, Thr	Asn, Asn, Lys, Lys	Ser, Ser, Arg, Arg
G Val, Val, Val, Val	Ala, Ala, Ala, Ala	Asp, Asp, Glu, Glu	Gly, Gly, Gly, Gly

(Each cell lists amino acids for 3rd base = U, C, A, G from top to bottom.)

Part 2: Transcription and Translation

DNA Template Strand: 3' - T A C G C A T T C A A C A T T - 5'

Step 1 — Transcribe: Write the mRNA by matching each DNA base to its RNA complement (A→U, T→A, G→C, C→G).

1. mRNA sequence: _____

Step 2 — Translate: Group the mRNA into codons (3 letters each). Use the codon table to find each amino acid.

2. Amino acid chain: _____

3. Which codon did you start with? What amino acid does it code for?

4. How did you know where to stop translating?

Part 3: Mutations

Go back to the original DNA template. Now imagine these changes:

Mutation A — Substitution: The 7th base (T) changes to a (C).

5. New mRNA: _____

6. New amino acids: _____

7. Did the protein change? Is this a silent, missense, or nonsense mutation?

Mutation B — Insertion: An extra A is inserted after the 3rd base.

8. New mRNA: _____

9. New amino acids: _____

10. Why did this insertion change so many amino acids? What is this type of mutation called?

Part 4: Summary

11. Fill in the blanks: DNA → ____ → ____

12. Where does transcription happen? ____

13. Where does translation happen? ____

14. What enzyme does transcription? ____

15. What structure does translation? ____